

```

{
  _id: 7,
  name: 'xReader',
  price: null
}
{
  _id: 5,
  name: 'SmartPhone',
  price: 599
}
{
  _id: 4,
  name: 'SmartPad',
  price: 699
}
db.products.insertMany([

```

```

  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },

```

```

  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },

```

```

  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },

```

```

  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
1024 ] },

```

```

  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },

```

```

  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] },

```

```

  { "_id" : 7, "name" : "xReader", "price" : null, "spec" : { "ram" : 64, "screen" :
6.7, "cpu" : 3.66 }, "color" : [ "black", "white" ], "storage" : [ 128 ] }

```

```

])

```

```
{
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5,
    '5': 6,
    '6': 7
  }
}
```

```
{
  _id: 1,
  name: 'xPhone',
  price: 799
}
{
  _id: 3,
  name: 'SmartTablet',
  price: 899
}
{
  _id: 2,
  name: 'xTablet',
  price: 899
}
```

```
db.products.find({
  'price': {
    $exists: 1
  }
}, {
  name: 1,
  price: 1
}).sort({
  price: 1,
  name: -1
})
```

```
{
  _id: 7,
  name: 'xReader',
```

```
    price: null
  }
  {
    _id: 5,
    name: 'SmartPhone',
    price: 599
  }
  {
    _id: 4,
    name: 'SmartPad',
    price: 699
  }
  {
    _id: 1,
    name: 'xPhone',
    price: 799
  }
  {
    _id: 2,
    name: 'xTablet',
    price: 899
  }
  {
    _id: 3,
    name: 'SmartTablet',
    price: 899
  }
}
```

```
db.products.find({
  releaseDate: {
    $exists: 1
  }
})
```

```
}, {
  name: 1,
  releaseDate: 1
}).sort({
  releaseDate: 1
});
```

```
{
  _id: 1,
  name: 'xPhone',
  releaseDate: 2011-05-14T00:00:00.000Z
}
{
```

```
  _id: 2,
  name: 'xTablet',
  releaseDate: 2011-09-01T00:00:00.000Z
}
{
  _id: 3,
  name: 'SmartTablet',
  releaseDate: 2015-01-14T00:00:00.000Z
}
{
  _id: 4,
  name: 'SmartPad',
  releaseDate: 2020-05-14T00:00:00.000Z
}
{
  _id: 5,
  name: 'SmartPhone',
  releaseDate: 2022-09-14T00:00:00.000Z
}
```

```
db.products.find({
  releaseDate: {
    $exists: 1
  }
})
```

```
}, {
  name: 1,
  releaseDate: 1
}).sort({
  releaseDate: -1
});
```

```
{
  _id: 5,
  name: 'SmartPhone',
  releaseDate: 2022-09-14T00:00:00.000Z
}
{
  _id: 4,
  name: 'SmartPad',
  releaseDate: 2020-05-14T00:00:00.000Z
}
{
  _id: 3,
  name: 'SmartTablet',
  releaseDate: 2015-01-14T00:00:00.000Z
}
```

```
}
{
  _id: 2,
  name: 'xTablet',
  releaseDate: 2011-09-01T00:00:00.000Z
}
{
  _id: 1,
  name: 'xPhone',
  releaseDate: 2011-05-14T00:00:00.000Z
}
```

```
db.products.find({}, {
  name: 1,
  spec: 1
}).sort({
  "spec.ram": 1
});
```

```
{
  _id: 1,
  name: 'xPhone',
  spec: {
    ram: 4,
    screen: 6.5,
    cpu: 2.66
  }
}
{
  _id: 5,
  name: 'SmartPhone',
  spec: {
    ram: 4,
    screen: 9.7,
    cpu: 1.66
  }
}
{
  _id: 4,
  name: 'SmartPad',
  spec: {
    ram: 8,
    screen: 9.7,
    cpu: 1.66
  }
}
```

```

{
  _id: 3,
  name: 'SmartTablet',
  spec: {
    ram: 12,
    screen: 9.7,
    cpu: 3.66
  }
}
{
  _id: 2,
  name: 'xTablet',
  spec: {
    ram: 16,
    screen: 9.5,
    cpu: 3.66
  }
}
{
  _id: 6,
  name: 'xWidget',
  spec: {
    ram: 64,
    screen: 9.7,
    cpu: 3.66
  }
}
{
  _id: 7,
  name: 'xReader',
  spec: {
    ram: 64,
    screen: 6.7,
    cpu: 3.66
  }
}

```

```

db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :

```

```

3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
1024 ] },
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] },
  { "_id" : 7, "name" : "xReader", "price" : null, "spec" : { "ram" : 64, "screen" :
6.7, "cpu" : 3.66 }, "color" : [ "black", "white" ], "storage" : [ 128 ] }
])

```

```

db.products.find({}, {
  name: 1,
  price: 1
}).sort({
  price: -1
}).limit(1);

{
  _id: 2,
  name: 'xTablet',
  price: 899
}

```

```

db.products.find({}, {
  name: 1,
  price: 1
}).sort({
  price: -1,
  name: 1
}).limit(1);

{
  _id: 3,
  name: 'SmartTablet',
  price: 899
}

```

```

db.products.find({}, {
  name: 1,
  price: 1
}

```

```
}).sort({
  price: -1,
  name: 1
}).skip(2).limit(2);
```

```
{
  _id: 1,
  name: 'xPhone',
  price: 799
}
{
  _id: 4,
  name: 'SmartPad',
  price: 699
}
```

```
db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
  ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
  2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
  ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
  3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
  ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
  3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
  ISODate("2020-05-14"),"spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":
  ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
])
```

```
db.products.updateOne({
  _id: 1
}, {
  $set: {
    price: 899
  }
})
```

```
{
  acknowledged: true,
```

```
insertedId: null,  
matchedCount: 1,  
modifiedCount: 1,  
upsertedCount: 0  
}
```

```
db.products.findOne({ _id: 1 }, { name: 1, price: 1 })
```

```
{  
  _id: 1,  
  name: 'xPhone',  
  price: 899  
}
```

```
db.products.find({ price: 899 }, { name: 1, price: 1 })
```

```
{  
  _id: 1,  
  name: 'xPhone',  
  price: 899  
}  
{  
  _id: 2,  
  name: 'xTablet',  
  price: 899  
}  
{  
  _id: 3,  
  name: 'SmartTablet',  
  price: 899  
}
```

```
db.products.updateOne({ price: 899 }, { $set: { price: null } })
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
db.products.find({ _id: 1 }, { name: 1, price: 1 })
```

```
{
  _id: 1,
  name: 'xPhone',
  price: null
}
```

```
db.products.find({ _id: 4 }, { name: 1, spec: 1 })
```

```
{
  _id: 4,
  name: 'SmartPad',
  spec: {
    ram: 8,
    screen: 9.7,
    cpu: 1.66
  }
}
```

```
db.products.updateOne({
  _id: 4
}, {
  $set: {
    "spec.ram": 16,
    "spec.screen": 10.7,
    "spec.cpu": 2.66
  }
})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
db.products.find({ _id: 4 }, { name: 1, spec: 1 })
```

```
{
  _id: 4,
  name: 'SmartPad',
  spec: {
    ram: 16,
```

```
    screen: 10.7,  
    cpu: 2.66  
  }  
}
```

```
db.products.updateOne(  
  { _id: 4 },  
  {  
    $set: {  
      "storage.0": 16,  
      "storage.1": 32  
    }  
  }  
)
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
db.products.find({ _id: 4 }, { name: 1, storage: 1 });
```

```
{  
  _id: 4,  
  name: 'SmartPad',  
  storage: [  
    16,  
    32,  
    1024  
  ]  
}
```

```
db.products.insertMany([  
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":  
    ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :  
    2.66 }, "color":["white","black"],"storage":[64,128,256]},  
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":  
    ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :  
    3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},  
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":  
    ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
```

```

3.66 }, "color": ["blue"], "storage": [16, 64, 128]},
    { "_id": 4, "name": "SmartPad", "price": 699, "releaseDate":
ISODate("2020-05-14"), "spec": { "ram": 8, "screen": 9.7, "cpu":
1.66 }, "color": ["white", "orange", "gold", "gray"], "storage": [128, 256, 1024]},
    { "_id": 5, "name": "SmartPhone", "price": 599, "releaseDate":
ISODate("2022-09-14"), "spec": { "ram": 4, "screen": 5.7, "cpu":
1.66 }, "color": ["white", "orange", "gold", "gray"], "storage": [128, 256]}
])

```

```

db.products.updateMany(
  { price: 899},
  { $set: { price: 895 }}
)

```

```

{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}

```

```

db.products.find({
  price: 895
}, {
  name: 1,
  price: 1
})

```

```

[
  { _id: 2, name: 'xTablet', price: 895 },
  { _id: 3, name: 'SmartTablet', price: 895 }
]

```

```

db.products.find({
  price: { $gt: 700}
}, {
  name: 1,
  price: 1,
  spec: 1
})

```

```

[
  {

```

```

    _id: 1,
    name: 'xPhone',
    price: 799,
    spec: { ram: 4, screen: 6.5, cpu: 2.66 }
  },
  {
    _id: 2,
    name: 'xTablet',
    price: 895,
    spec: { ram: 16, screen: 9.5, cpu: 3.66 }
  },
  {
    _id: 3,
    name: 'SmartTablet',
    price: 895,
    spec: { ram: 12, screen: 9.7, cpu: 3.66 }
  }
]

```

```

db.products.updateMany({
  price: { $gt: 700}
}, {
  $set: {
    "spec.ram": 32,
    "spec.screen": 9.8,
    "spec.cpu": 5.66
  }
})

```

```

{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}

```

```

db.products.updateMany({
  _id: {
    $in: [1, 2, 3]
  }
}, {
  $set: {
    "storage.0": 16,
    "storage.1": 32
  }
})

```

```
}  
})
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 3,  
  modifiedCount: 3,  
  upsertedCount: 0  
}
```

```
db.products.find(  
  { _id: { $in: [1,2,3]}},  
  { name: 1, storage: 1}  
)
```

```
{  
  _id: 1,  
  name: 'xPhone',  
  storage: [  
    64,  
    128,  
    256  
  ]  
}  
{  
  _id: 2,  
  name: 'xTablet',  
  storage: [  
    128,  
    256,  
    512  
  ]  
}  
{  
  _id: 3,  
  name: 'SmartTablet',  
  storage: [  
    16,  
    64,  
    128  
  ]  
}
```

```

db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
  ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
  2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
  ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
  3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
  ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
  3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
  ISODate("2020-05-14"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":
  ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
])

```

```

db.products.updateOne({
  _id: 1
}, {
  $inc: {
    price: 50
  }
})

```

```

{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

```

db.products.find(
  { _id: 1},
  {name: 1, price: 1}
)

```

```

[ { _id: 1, name: 'xPhone', price: 849 } ]

```

```

db.products.updateOne({
  _id: 1
}, {

```

```
    $inc: {
      price: -150
    }
  })
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
db.products.find(
  { _id: 1 },
  { name: 1, price: 1 }
)
```

```
[ { _id: 1, name: 'xPhone', price: 699 } ]
```

```
db.products.updateOne({
  _id: 1
}, {
  $inc: {
    price: 50,
    "spec.ram": 4
  }
})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
db.products.find(
  { _id: 1 },
  { name: 1, price: 1, "spec.ram": 1 }
)
```

```
[ { _id: 1, name: 'xPhone', price: 749, spec: { ram: 8 } } ]
```

```

db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
  ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
  2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
  ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
  3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
  ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
  3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
  ISODate("2020-05-14"),"spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":
  ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
])

```

```

db.products.updateOne({
  _id: 5
}, {
  $min: {
    price: 699
  }
})

```

```

{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}

```

```

db.products.updateOne({
  _id: 5
}, {
  $min: {
    price: 499
  }
})
{

```

```
    acknowledged: true,  
    insertedId: null,  
    matchedCount: 1,  
    modifiedCount: 1,  
    upsertedCount: 0  
  }
```

```
db.products.find({ _id: 5 }, { name: 1, price: 1 })
```

```
[ { _id: 5, name: 'SmartPhone', price: 499 } ]
```

```
db.products.insertMany([  
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":  
    ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :  
    2.66 }, "color":["white","black"],"storage":[64,128,256]},  
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":  
    ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :  
    3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},  
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":  
    ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :  
    3.66 }, "color":["blue"],"storage":[16,64,128]},  
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":  
    ISODate("2020-05-14"),"spec" : { "ram" : 8, "screen" : 9.7, "cpu" :  
    1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},  
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":  
    ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :  
    1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}  
])
```

```
db.products.updateOne({  
  _id: 1  
}, {  
  $max: {  
    price: 699  
  }  
})
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

```
}
```

```
db.products.updateOne({  
  _id: 1  
}, {  
  $max: {  
    price: 899  
  }  
})
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
db.products.find({ _id: 1 }, { name: 1, price: 1 })
```

```
[ { _id: 1, name: 'xPhone', price: 899 } ]
```